



ENHANCING DAIRY RETAIL OPERATIONS THROUGH ARTIFICIAL INTELLIGENCE AND ADVANCED ANALYTICS

Prepared by:

LEE SHIEN YING (25085358)

JASRINA KAUR SIDHU (25058181)

JASPREET KAUR GILL (25084054)

INTRODUCTION

- Dairy retail involves highly perishable, short shelf-life products.
- Stockouts and overstocking cause sales loss and spoilage.
- Dataset offers real-world insights on sales, pricing, stock, expiry, and customer behaviour.
- Traditional manual planning is insufficient for fast-changing demand.
- AI supports better forecasting, expiry management, and pricing decisions.



LITERATURE REVIEW



LITERATURE SHOWS AI IS ESSENTIAL FOR IMPROVING FORECASTING, INVENTORY CONTROL, AND PRICING FOR PERISHABLE DAIRY PRODUCTS.

01

Perishable Goods Demand Forecasting in Retail

Advanced time-series and segment-based models improve accuracy for volatile demand.

(Zhu, 2024; Jauhar et al., 2024)

02

Perishable Inventory and Expiry Management

AI-driven predictive models reduce spoilage and support better replenishment.

(Birkmaier et al., 2024; Korie, 2025)

03

Pricing Strategy and Revenue Optimization for FMCG

Dynamic pricing guided by forecasting improves profitability and reduces waste.

(Korie, 2025; Komijani & Sajadieh, 2024)

04

Role of AI and Data-Driven Decision Support in Retail

AI and digital twins enhance operational visibility and real-time decision-making.

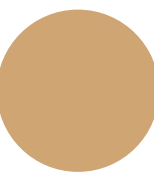
(Falayi et al., 2024; Maheshwari et al., 2023)

PROBLEMS STATEMENT

CATEGORY	CHALLENGE	CONSEQUENCE
Challenges in Managing Perishable Dairy Goods	Siloed decision-making fails to handle short shelf life and high demand volatility.	Causes frequent stockouts, overstocking, and spoilage due to poor handling of rapid fluctuations.
Limitations in Inventory and Expiry Management	Static stock levels and manual rotation are inadequate for real-time spoilage monitoring.	Leads to significant avoidable food waste and inefficient cold-chain performance.
Gaps in Pricing Strategies for Perishable Goods	Dynamic pricing is poorly integrated with broader forecasting and inventory frameworks.	Results in fragmented decision-making, missing opportunities for optimal revenue and waste reduction.
Limited Adoption of AI and Predictive Analytics	Adoption is limited by data constraints, interpretability issues, and integration complexities.	Hinders the industry from leveraging advanced tools for holistic supply chain optimization.
Research Gap and Project Focus	Current methods are inadequate for the scale and complexity of perishable dairy management.	An urgent need exists for an integrated, AI-driven decision-support tool tailored for this domain.



DATA PREPARATION



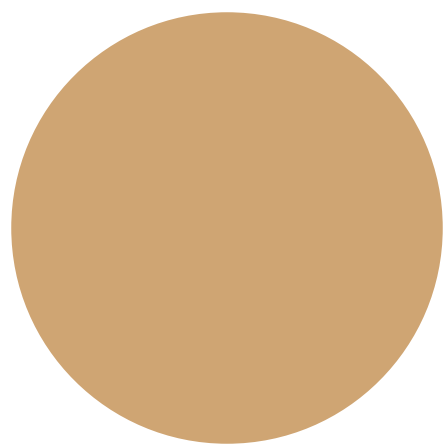
Data Processing Steps

- Reformatted Date, Production Date, and Expiration Date into consistent datetime format
- Corrected unreadable or invalid dates to prevent errors during modelling
- Created a new target feature:
 - → LowStock = 1 when stock < threshold
 - → 0 otherwise

Feature Engineering & Encoding

- Numeric features included production volume, shelf life, revenue, and stock levels
- Categorical features included product name, brand, storage condition, and location
- Applied StandardScaler to normalize numeric values
- Applied OneHotEncoder to convert categorical variables into binary format

Train-Test Split

- 80:20 split
 - Stratified sampling used to maintain class balance for the LowStock target
- 



AI MODEL DEVELOPMENT

Three Machine Learning Models Developed

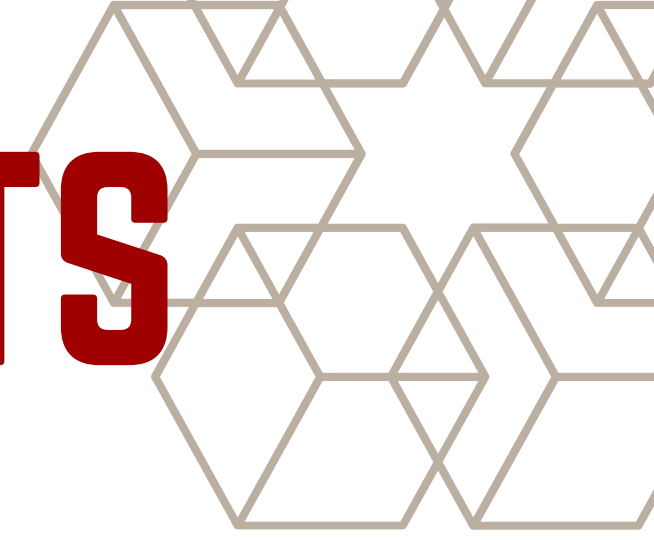
MODEL	PURPOSE	METHOD
Demand Forecasting	Identify demand trends and support production planning	7-Day Moving Average (time-series based)
Low-Stock Classification	Predict items at risk of shortage	Logistic Regression
Pricing Prediction	Estimate price per unit based on product and operational factors	Linear Regression

Development Workflow

- Preprocessing using ColumnTransformer
- Model trained on cleaned dataset using Scikit-learn
- Performance evaluated using relevant metrics based on task



MODEL RESULTS & OUTPUTS



Key Model Evaluations

- Demand forecasting model captures overall demand trends using a 7-day moving average
- Low-stock classification model shows high recall and strong accuracy
- Confusion matrix shows minimal false negatives
- ROC-AUC score close to 1 indicates strong predictive power
- Pricing model predictions closely align with actual values based on scatterplot
- MAE, RMSE and R^2 metrics confirm strong model performance

How the Outputs Support Decision-Making

- Forecasting insights help plan production efficiently
- Low-stock predictions ensure timely replenishment
- Pricing recommendations help maximise revenue and reduce losses



MODEL DEPLOYMENT (GRADIO WEB APP)



Model Deployment (Gradio Web App)

- Built a simple interactive web application using Gradio
- Allows users to:
 - Predict low stock risk
 - Estimate optimal selling price
 - Forecast future demand
- Application generates a public web link for real-world usability

This demonstrates how the AI solution can be applied in operational decision-making.



CONCLUSION

- Successfully developed and implemented a unified AI framework linking Forecasting, Inventory Control, and Pricing.
- Models provide reliable, data-driven outputs for proactive replenishment (reducing stockouts) and strategic pricing (optimizing revenue).
- The framework achieves the critical dual goal for perishable retail:
 - Waste Reduction: Proactive expiry and stock management minimizes financial and environmental loss.
 - Profitability: Optimized inventory levels and dynamic pricing maximize returns.
- Transitioned the supply chain from rigid, historical planning to a flexible, forecast-driven model, vital for volatile FMCG environments.
- Digital transformation is a strategic necessity for ensuring long-term sustainability and resilience in the high-pressure dairy sector.



THANK YOU